

Large-polaron transport

10.1 Coherent versus incoherent transport

Electronic transport is often described in terms of two complementary models. Coherent transport envisions a carrier moving through a material with its motion occasionally disrupted by scattering events. By contrast, incoherent transport views a carrier as only making occasional jumps between localized regions. Coherent transport in conventional semiconductors is generally ascribed to a carrier with a mobility that greatly exceeds $1 \text{ cm}^2/\text{V}\cdot\text{sec}$ and falls with increasing temperature. Incoherent transport is often attributed to a carrier with a mobility that is much less than $1 \text{ cm}^2/\text{V}\cdot\text{sec}$ and rises with increasing temperature.

Treatments of coherent transport usually begin with a charge carrier in a delocalized state whereas incoherent transport starts with a carrier in a localized state. As a result, delocalized states are often said to produce coherent transport while localized states produce hopping transport. However, there is not a one-to-one correspondence between the nature of a carrier's eigenstates and its transport. The character of a carrier's transport also depends on its scattering. For example, periodicity ensures that the eigenstates of a small polaron in a crystal are Bloch-like delocalized states. Nonetheless, as will be discussed in Chapter 11, small-polaron transport typically proceeds by phonon-assisted hopping. Coherent transport is then destroyed as scattering reduces a small-polaron's mean-free-path ℓ to below its diameter $d_c < a$, the lattice constant: $\ell < d_c$. Thus, transport among delocalized eigenstates can be incoherent. In addition, transport among localized states can be coherent. For example, consider the Wannier–Stark localized electronic eigenstates formed when an electric field F is applied across a periodic potential of lattice constant a (Wannier, 1959; Emin and Hart, 1987a). The eigenvalues comprise a Wannier–Stark ladder whose successive rungs are separated in energy by eFa . The corresponding electronic eigenstates are each localized with the Wannier–Stark length of $L_{\text{W-S}} \sim W/eF$ about successive lattice

sites, where W is the electronic bandwidth in the absence of the applied electric field. With $L_{W-S} \gg a$, many Wannier–Stark eigenstates strongly overlap one another even as their spatial oscillations ensure their mutual orthogonality. However, scattering mixes these states. Coherent transport results when the Wannier–Stark length exceeds the carrier mean-free-path, $L_{W-S} > \ell$ (Kane, 1959; Emin and Hart, 1987b). Thus, conventional semiconductors display coherent transport as evidence of localization is obliterated. By contrast, Wannier–Stark localization is indicated when $L_{W-S} \ll \ell$. This exceptional circumstance arises when a sufficiently strong electric field is applied to a narrow-band super-lattice whose carriers have a long mean-free-path (Mendez *et al.*, 1988).

In summary, coherent scattering-type transport can be envisioned when a carrier's mean-free-path exceeds its diameter d_c . Coherent transport even occurs among localized states provided that the carrier's mean-free-path is sufficient to disrupt the localization: $L_{loc} > \ell > d_c$. Incoherent transport occurs when scattering destroys the carrier's integrity: $\ell < d_c$. The forthcoming discussion will develop these notions further.

The Boltzmann equation is frequently used to derive an expression for a carrier's mobility μ in terms of its velocity v and mean-free-path ℓ . This relationship can be used to establish the minimum mobility for coherent transport. Specifically, this lower limit is found by replacing a carrier's mean-free-path by its characteristic length, d_c . It is then noted that a coherently moving carrier's diameter is related to its de Broglie wavelength λ : $d_c \equiv \lambda/2\pi = \hbar/p = \hbar/m^*v$, where p and m^* are the carrier's momentum and mass, respectively. The carrier mobility μ for scattering-type coherent transport is constrained by:

$$\mu = \frac{e}{\kappa T} v \ell > \frac{e}{\kappa T} v d_c = \frac{e}{\kappa T} \frac{\hbar}{m^*}. \quad (10.1)$$

Thus, for a conventional wide-band semiconductor, self-consistency of the standard Boltzmann-equation-based scattering treatment requires a high mobility (Fröhlich and Sewell, 1959; Herring, 1961). For example, the requirement of Eq. (10.1) evaluated at 300 K with m^* even as large as the free-electron mass is that $\mu > 45 \text{ cm}^2/\text{V-sec}$. However, the Boltzmann-scattering approach can be applied to a much lower mobility if the carrier's effective mass is enhanced significantly above the free-electron mass. Such large effective masses occur with large-polaron formation. Room-temperature Hall mobilities of $\sim 1\text{--}10 \text{ cm}^2/\text{V-sec}$ that are reported for n -type carriers in alkali halides, alkaline-earth fluorides and some transition-metal oxides have been attributed to large polarons (Seager and Emin, 1970; Seager, 1971; Forro *et al.*, 1994).

10.2 Large-polaron effective mass

A self-trapped carrier can only move when the atoms responsible for its binding alter their positions. This movement of atoms generally changes the self-trapped carrier's potential well and therefore its energy. If the variation of the self-trapped carrier's energy during its motion remains less than its electronic band half-width $W/2$ (the energetic measure of its coherence) its motion is coherent.

The motion of a large-polaron's self-trapped carrier generally retains its coherence. In particular, in accord with Section 4.3, the energy of the self-trapped carrier of a large polaron is $E_{\text{st}} = -V_{\text{def}}^2/W$, where V_{def} is the deformation-related energy and the condition $W/2 > V_{\text{def}}$ insures that the polaron is large (Holstein, 1959a). The change of the self-trapped carrier's energy associated with its motion is $\Delta E_{\text{st}} = 2V_{\text{def}}\Delta V_{\text{def}}/W$, where ΔV_{def} is the change of its deformation-related energy associated with moving the carrier a lattice constant. Large-polaron motion will be coherent if $1 > |\Delta E_{\text{st}}|/(W/2) = V_{\text{def}}\Delta V_{\text{def}}/(W/2)^2$. This condition is fulfilled since large-polaron formation and motion is generally conditioned by $W/2 > V_{\text{def}} \gg \Delta V_{\text{def}}$.

A large polaron's effective mass is often defined as the net shift of atomic mass associated with translating an equilibrated self-trapped carrier by a lattice constant. In particular, components of the effective-mass tensor $M_{j,j'}$ link the kinetic energy of a large polaron with components of its velocity in directions j and j' . That is, the large-polaron kinetic energy is

$$\text{K.E.} \equiv \frac{1}{2} \sum_{j=1}^3 \sum_{j'=1}^3 M_{j,j'} \left(\frac{\delta x_j}{\delta t} \right) \left(\frac{\delta x_{j'}}{\delta t} \right), \quad (10.2)$$

where δx_j is the distance that the self-trapped carrier is translated in direction j during time interval δt . Here the effective-mass tensor is defined by

$$M_{j,j'} \equiv \sum_{\mathbf{g}} \sum_{i=1}^3 M_{\mathbf{g}} \left(\frac{\delta u_{\mathbf{g},i}}{\delta x_j} \right) \left(\frac{\delta u_{\mathbf{g},i}}{\delta x_{j'}} \right), \quad (10.3)$$

where $\delta u_{\mathbf{g},i}$ is the shift in direction i of the atom of mass $M_{\mathbf{g}}$ located at site $\mathbf{g}$ that moves the carrier δx_j in direction j and $\delta x_{j'}$ in direction j' . The shifts of atoms' equilibrium displacements after translating the carrier by a lattice constant in the j -direction equals the differences between equilibrium displacements in the negative j -direction of adjacent pairs of atoms: $\delta u_{\mathbf{g},i} \rightarrow \delta U_{\mathbf{g},i}$. Thus, the large-polaron's effective-mass tensor can be rewritten in terms of its static properties as:

$$M_{j,j'} = \sum_{\mathbf{g}} \sum_{i=1}^3 M_{\mathbf{g}} \left(\frac{\delta U_{\mathbf{g},i}}{\delta a_j} \right) \left(\frac{\delta U_{\mathbf{g},i}}{\delta a_{j'}} \right), \quad (10.4)$$

where the two bracketed terms are elements of the polaron's carrier-induced strain tensor with δa_j denoting the lattice constant in the j -direction.

Equation (10.4) has been evaluated for different models of the electron–phonon interaction (Emin and Hillery, 1989). For the three-dimensional large-polaron formed with the Fröhlich long-range interaction the effective mass is [cf. Eq. (4.39)]:

$$m_{\text{LP}} \approx \frac{E_{\text{LP}}}{\omega^2 R_{\text{LP}}^2}, \quad (10.5)$$

where E_{LP} and R_{LP} are the large-polaron binding energy and radius, respectively, and ω is the optical-vibration frequency. The three-dimensional result has also been expressed in terms of the Fröhlich coupling constant α as $m\alpha^4$ with $\alpha \gg 1$, where m is the free-carrier mass and α is defined in Eq. (3.3) (Fröhlich, 1963). For a one-dimensional large polaron formed by a carrier's interaction with acoustic phonons the effective mass is $\sim 4E_{\text{LP}}/c_s^2$, where c_s is the sound velocity (Schüttler and Holstein, 1986). These formulae indicate that large-polaron effective masses are typically orders of magnitude greater than the free-electron mass.

A large polaron moves very slowly with a very large mass because its self-trapped carrier only moves when the atoms associated with its self-trapping alter their positions. Concomitantly, the large mass of a large polaron can only be observed by measurements performed at frequencies Ω well below that of vibrating atoms: $\Omega \ll \omega$ (e.g. at microwave frequencies). Higher-frequency measurements observe just the mass of the self-trapped electron not the mass of the large polaron. That is, such measurements primarily observe electronic motion within the self-trapped well.

The preceding discussion addressed the adiabatic effective mass of a large polaron associated with atoms' continuous classical motion. However, a second type of large-polaron motion is possible. The self-trapped carrier can follow atoms as they tunnel between discrete geometrically equivalent atomic-deformation patterns (cf. Fig. 10.1). By itself, such tunneling motion is coherent since it involves no energetic variations. However, atomic tunneling significantly reduces the coherence

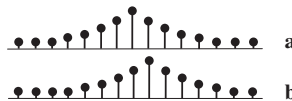


Fig. 10.1 Displacements of atoms' equilibrium positions are schematically illustrated for a one-dimensional large polaron before (a) and after (b) it is translated by one lattice constant to the right. The disparity between an atom's displacement before and after translation is the same as that between the atom and its nearest-neighbor to its left.

energy (the half-bandwidth) for this coordinated electronic and atomic motion from that for pure electronic motion. Thus, this type of coherent transport will be squelched with sufficient energetic disorder and scattering.

The bandwidth for such large-polaron motion is proportional to the matrix element of the system's adiabatic Hamiltonian between atomic-vibration wavefunctions associated with large polarons centered at different sites. Beyond having different centroids these two wavefunctions represent geometrically equivalent configurations of atomic equilibrium displacements and equal quantum numbers. The adiabatic Hamiltonian describes a self-trapped carrier among a periodic array of vibrating atoms. In accord with the adiabatic principle these atoms' equilibrium positions change in concert with the location of the self-trapped carrier. Thus, a large polaron's adiabatic potential comprises a periodic array of deformable potential wells. All told, the bandwidth associated with a large polaron tunneling between states with different centroids is a generalization of adiabatic small-polaron tunneling between the two sites of a double well (Holstein, 1959b).

Atoms' vibrations about their equilibrium positions are described by the product of displaced-oscillator wavefunctions. Translating atomic equilibrium positions of the displaced-oscillator wavefunctions coherently shifts a self-trapped carrier between geometrically equivalent positions. The overlap between the products of displaced-harmonic-oscillator wavefunctions that correspond to a self-trapped carrier shifted by the vector $\mathbf{h}$ is defined as $\exp[-S(\mathbf{h})]$, where the vibrations' quantum numbers have been replaced by their thermal average (Holstein, 1959b).

For definitiveness, consider large-polaron formation on a linear chain of displaceable atoms labeled by the index g that vibrate about the equilibrium displacement $u_{p,g}$ with the frequency ω and stiffness k when the self-trapped carrier is centered at site p (Holstein, 1959a). The overlap function for this model is

$$e^{-S(h)} = \exp \left[- \sum_{g=-\infty}^{\infty} \frac{k(u_{p,g} - u_{p,g-h})^2}{4\hbar\omega} \coth(\hbar\omega/2\kappa T) \right], \quad (10.6)$$

where the symmetry condition $u_{p+h,g} = u_{p,g-h}$ has been used. This overlap factor vanishes in the classical limit, $\hbar \rightarrow 0$, where atomic tunneling is prohibited. The overlap factor also falls with increasing temperature since the overlap between displaced harmonic-oscillator wavefunctions decreases with increasing vibration energy as additional nodes produce more cancellations. Finally, the overlap falls rapidly as h , the separation between the centroids of the overlapping large-polarons, is increased.

The nearest-neighbor transfer energy for this one-dimensional large-polaron is

$$t_{\text{LP}} \approx \sqrt{\frac{3E_{\text{LP}}(L)\hbar\omega}{\pi L}} \exp\left[-\frac{E_{\text{LP}}(L)}{\hbar\omega L^2} \coth(\hbar\omega/2\kappa T)\right], \quad (10.7)$$

where L is the half-length of the large polaron expressed in units of the lattice constant and $E_{\text{LP}}(L) \propto 1/L$ is the large-polaron binding energy derived by Holstein (1959a). With pure nearest-neighbor transfer the one-dimensional large-polaron bandwidth from this tunneling motion is $W_{\text{LP,tun}} = 4t_{\text{LP}}$. The width of this large-polaron tunneling band is typically less than the phonon energy $W_{\text{LP,tun}} < \hbar\omega$ given the requirement for large-polaron formation $E_{\text{LP}}(L) \gg \hbar\omega$.

Distinctively, this tunneling bandwidth narrows with increasing temperature. This narrowing becomes most pronounced once κT rises above $\hbar\omega/2$. Then the large-polaron effective mass associated with tunneling motion, $\propto 1/W_{\text{LP,tun}}$, rises exponentially with increasing temperature. This feature has been suggested as a possible explanation of the steep fall of the large-polaron Hall mobility as the temperature is raised above the characteristic phonon temperatures of alkali halide and alkaline-earth fluoride crystals (Seager and Emin, 1970; Seager, 1971).

10.3 Large-polaron scattering

The dynamics of a large-polaron's motion are qualitatively different from those of an electronic charge carrier in a conventional semiconductor. For definiteness consider a one-dimensional large polaron which interacts with acoustic phonons (Schüttler and Holstein, 1986). The ratio of a large polaron's mass m_{LP} to that of a conventional electronic carrier m is huge: $\sim W_{\text{e}}E_{\text{LP}}/(\hbar\omega_{\text{D}})^2 \gg 1$, where W_{e} is the conventional-carrier's electronic bandwidth, E_{LP} is the large polaron binding energy and ω_{D} is the Debye frequency. Furthermore, the thermal velocity of a large polaron is much less than that of a conventional carrier: $\sim (m/m_{\text{LP}})^{1/2}$. The large polaron's thermal velocity is even smaller than the sound velocity c_{s} : $\sim c_{\text{s}}(\kappa T/E_{\text{LP}})^{1/2}$ with $E_{\text{LP}} \gg \kappa T$. Moreover, the thermal momentum of a large polaron p_{th} is much larger than that of a conventional electronic carrier: $\sim (m_{\text{LP}}/m)^{1/2}$. The large polaron's thermal momentum even exceeds the thermal momentum of an acoustic phonon $\hbar k_{\text{th}} \equiv \kappa T/c_{\text{s}}$ by $(E_{\text{LP}}/\kappa T)^{1/2}$.

These dynamic considerations depict a large polaron moving slowly with a large momentum while scattering with phonons having relatively small momenta. By contrast, an electronic carrier in a conventional semiconductor usually moves rapidly with small momentum while scattering against phonons with relatively large momenta. For these reasons large polarons are only weakly scattered by phonons while conventional electronic carriers are strongly scattered by phonons.

A large polaron's scattering with phonons occurs as they encounter the region of softened vibrations that surrounds a self-trapped carrier. As depicted in Fig. 10.2, a self-trapped carrier adjusts to alterations of the positions of the atoms associated with its self-trapping. This electronic relaxation reduces the net energy expended in displacing the associated atoms thereby reducing their stiffness constants. Affected stiffness constants can be reduced by significant fractions [estimates may be made using Eqs. (4.15) and (4.18)]. The cross-section for a large polaron's scattering by phonons peaks at the resonance when the phonon wavevector is comparable to the reciprocal of the softened region's spatial extent (Schüttler and Holstein, 1986).

A large-polaron's scattering by phonons is unlike that of a conventional carrier. A large-polaron's huge effective mass precludes most processes in which a single phonon is emitted or absorbed since $m_{LP}c_s \gg p_{th} \gg \hbar k_{th}$. Thus a large-polaron's scattering is dominated by processes in which relatively little net energy is transferred between the phonons and the large polaron. By contrast, the width of the carrier's energy band in a conventional semiconductor is very much greater than phonon energies. Therefore scattering of a conventional carrier occurs primarily via the absorption or emission of single phonons.

The predominant process by which a large polaron is scattered by ambient phonons is illustrated in Fig. 10.3. A slow-moving large-momentum large polaron is scattered as it "reflects" faster-moving low-momentum phonons (Schüttler and Holstein, 1986). This process is analogous to that by which a macroscopic ball moving through air encounters resistance by colliding with relatively light fast-moving molecules.

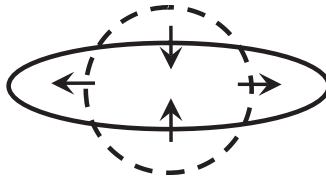


Fig. 10.2 A self-trapped carrier (dashed circle) alters its shape (to form an oval) in response to atoms' movements. The self-trapped carrier's polarization (emphasized with arrows) reduces the net stiffness governing these atomic displacements.

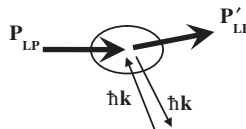


Fig. 10.3 The softened region of a large polaron (the ellipse) scatters incident phonons. The large momentum of a thermal heavy-massed large polaron enables it to reflect the relatively low-momentum phonons it scatters.

Consider how a one-dimensional large polaron of length L is scattered by ambient acoustic phonons indexed by wavevector k (Schüttler and Holstein, 1986). The large polaron's scattering rate is the thermal average of the product of (1) the density of ambient phonons $N(k/k_{\text{th}})$, (2) the cross-section for their being reflected by the large polaron $\sigma(kL)$, and (3) the resulting change of a large polaron's velocity $2\hbar k/m_{\text{LP}}$:

$$\frac{1}{\tau} = \left\langle N(k/k_{\text{th}}) \sigma(kL) \left(\frac{2\hbar k}{m_{\text{LP}}} \right) \right\rangle_{\text{thermal}}, \quad (10.8)$$

where it is noted that a phonon transfers twice its momentum to a polaron from which it is reflected and $\hbar k_{\text{th}} \equiv \kappa T/c_s$ is the thermal-acoustic-phonon momentum. In the low-temperature limit the phonons' Bose factor $\approx \exp(-k/k_{\text{th}})$ restricts their scattering to those for which $kL \ll 1$ with $\sigma(kL) \propto (kL)^2$. The mobility then becomes

$$\mu_{\text{LP}} = \frac{e\tau}{m_{\text{LP}}} \propto \frac{eL^2}{\hbar} \left(\frac{T_L}{T} \right)^4, \quad (10.9)$$

where $\kappa T_L \equiv \hbar c_s/L \gg \kappa T$. In the regime $T \gg T_L$ the phonons' Bose factor $\approx k_{\text{th}}/k$ permits their scattering to be dominated by those for which $kL \approx 1$ with $\sigma(kL) \approx 1$ yielding

$$\mu_{\text{LP}} \approx \frac{eL^2}{\hbar} \left(\frac{T_L}{T} \right) = \frac{ec_s L}{\kappa T}. \quad (10.10)$$

Acoustic phonons' weak scattering of a one-dimensional large polaron, indicated by its long scattering time, confirm that its transport is coherent. In particular, even when $T > T_L$ the large-polaron's mean-free-path exceeds its spatial extent:

$$\ell = v_{\text{LP}} \tau \approx \left(\frac{2\kappa T}{m_{\text{LP}}} \right)^{1/2} \left(\frac{m_{\text{LP}} c_s L}{\kappa T} \right) = \left(\frac{8E_{\text{LP}}}{\kappa T} \right)^{1/2} L \gg L, \quad (10.11)$$

where it is noted that $\tau = m_{\text{LP}} c_s L / \kappa T$, cf. Eq. (10.10), and recalled that $m_{\text{LP}} \approx 4E_{\text{LP}}/c_s^2$.

The preceding discussion of this section has the large-polaron's very long scattering time compensating for its very big effective mass ($\tau \propto m_{\text{LP}}$) to produce a mobility that is independent of its effective mass. In fact, the magnitude of this large-polaron's mobility is comparable to that of conventional charge carriers: $> 1\text{--}10 \text{ cm}^2/\text{V}\cdot\text{sec}$ at 300 K. In essence, the weak phonon scattering of this slow-moving massive large polaron gives a similar mobility to that from strong phonon scattering of a fast-moving conventional electronic carrier.

Distinctive aspects of this large-polaron's transport may nonetheless be observed. The unconventionally long large-polaron scattering time shifts the Drude tail of the frequency-dependent conductivity $\text{Re}[\sigma(\Omega)] \propto 1/[1 + (\Omega\tau)^2]$ to exceptionally low frequencies, toward the microwave regime [cf. Section 9.2]. Furthermore, the scattering of acoustic phonons by large polarons is unlike phonons' absorption and emission by conventional carriers. This distinctive feature may be discernable from studying the effect of carriers on phonons' thermal diffusivity.

Large-polaron transport is dramatically different for scattering with dispersion-less optic phonons (Holstein, 1981; Schüttler and Holstein, 1986). In the absence of optic-phonons' dispersion their group velocity vanishes. Then optic phonons behave as thermally generated stationary imperfections which scatter the large polaron. The mobility is inversely proportional to the density of these scattering centers. Thus the mobility garners a factor which falls with increasing temperature as $\exp(\hbar\omega/\kappa T)$ when $\kappa T \ll \hbar\omega$, the optic-phonon energy. Moreover, a large-polaron's scattering cross-section with dispersion-less optical phonons falls once its momentum exceeds $\hbar/L$ (Holstein, 1981; Schüttler and Holstein, 1986). This effect contributes a factor to the mobility that rises with increasing temperature as $\exp(\pi^2\kappa T/E_c)$ for $\kappa T \gg E_c$, the kinetic energy at which a large polaron's de Broglie wavelength $\hbar/p_{\text{LP}}$ equals its length L : $(p_{\text{LP}})^2/2m_{\text{LP}} = (\hbar/L)^2/2m_{\text{LP}} \equiv E_c \ll \hbar\omega$ [cf. Eq. (4.40)]. Thus, the mobility's temperature dependence results from a competition between (1) the number of scattering centers that increases with rising temperature and (2) the efficacy of these centers' scattering that falls with increasing temperature. The first factor dominates at lower temperatures to produce a mobility that falls with increasing temperature until $\kappa T = (E_c\hbar\omega)^{1/2}/\pi$. As the temperature is increased further the second effect begins to dominate causing the mobility to rise from its minimum value.

All told, a cogent view of a large polaron's scattering with phonons has emerged. Polarization of a large-polaron's self-trapped carrier in response to atoms' movements softens their vibrations. The softened region then scatters phonons. The nature of the resulting scattering depends critically on the large-polaron's velocity $(2\kappa T/m_{\text{LP}})^{1/2}$ relative to the phonons' group velocity $\partial\omega(k)/\partial k$. In one limit the relatively slow moving large polaron reflects faster moving acoustic phonons that impinge on it. In the complementary limit the large polaron is scattered by stationary optic phonons. That is, phonons' dispersion determines whether quasi-particles' relative motion is attributed primarily to phonon wavepackets or to large polarons. The scattering peaks when the wavelength of the faster quasi-particle coincides with the spatial extent of the self-trapped carrier. Beyond these idealized models self-trapped electrons generally interact with both acoustic and optic modes. Furthermore optic modes often possess significant dispersion.